

R A G
Retrieval Augmented Generation

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We will give a live demo of RAG

01

Theory lesson



Introduction to RAG

Retrieval-Augmented Generation (RAG) is an AI framework that connects Large Language Models (LLMs) to external, private knowledge sources.

Instead of just relying on what the AI learned during its initial training, RAG lets the model "look up" real-time or proprietary information to provide highly accurate, context-aware answers.



Why do we need RAG?

Limitation of LLMS

When asked about specific company data, an LLM will resort to guesswork either inventing plausible-sounding details (**hallucinating**) or falling back on **generic industry data**.

Zero-Shot Learning

RAG gives the model instant access to live data streams. You get up-to-the-minute accuracy with zero retraining costs.



How RAG helps



Fixing the Company Data Gap

RAG searches your private database for relevant document snippets and attaches them to the prompt, forcing the LLM to answer using your verified corporate facts instead of generic guesses.



Enabling "Zero-Shot" Live Updates

Venus has a beautiful name and is the second planet from the Sun. It's hot and has a poisonous atmosphere

02

RAG Pipeline



RAG pipeline



Document Injection

Mercury is the closest planet to the Sun and the smallest of them all



Information retrieval

Venus has a beautiful name and is the second planet from the Sun



Response Generation

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

Information Extraction Process

LOAD



Loading the data that we want to apply RAG on.

SPLIT



Chunking that is splitting the data because of Token Limit

EMBED

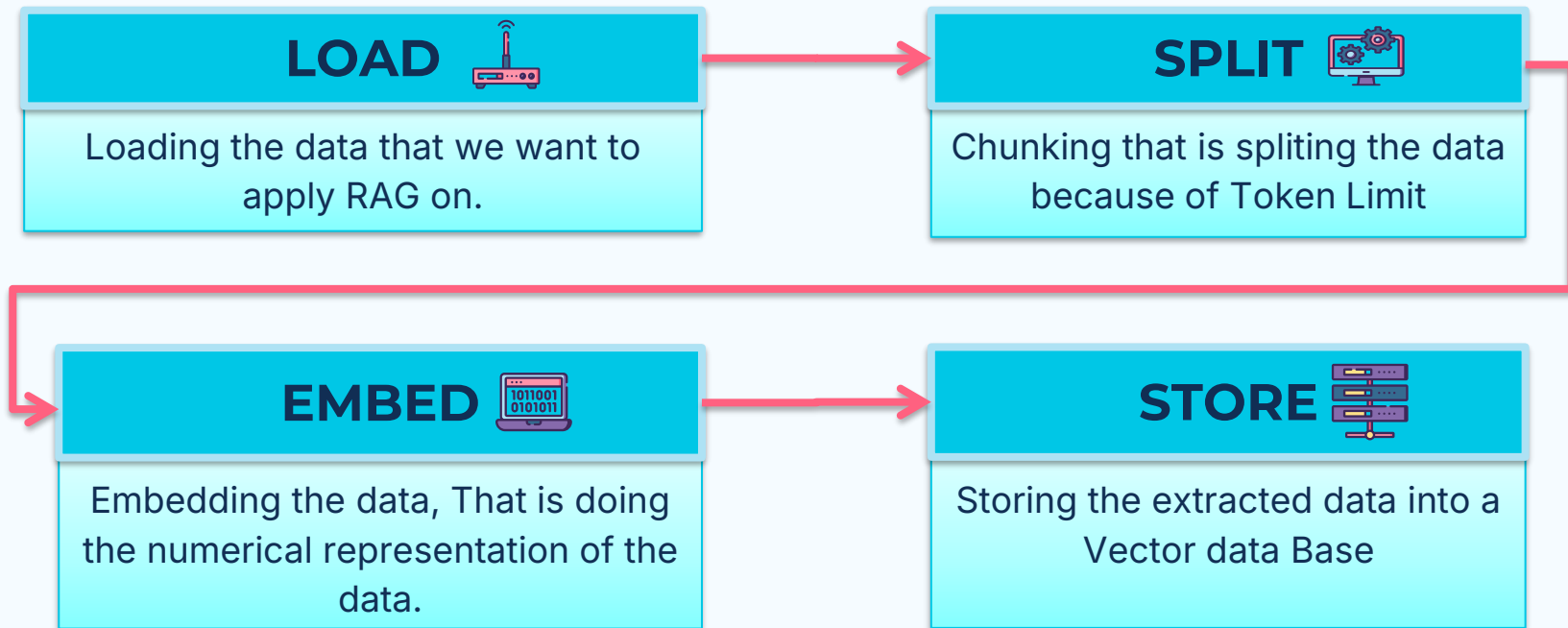


Embedding the data, That is doing the numerical representation of the data.

STORE



Storing the extracted data into a Vector data Base



Information Retrival Process

Question



The user enters a specific prompt or query. This initiates the process, defining exactly what information needs to be found.



STORE

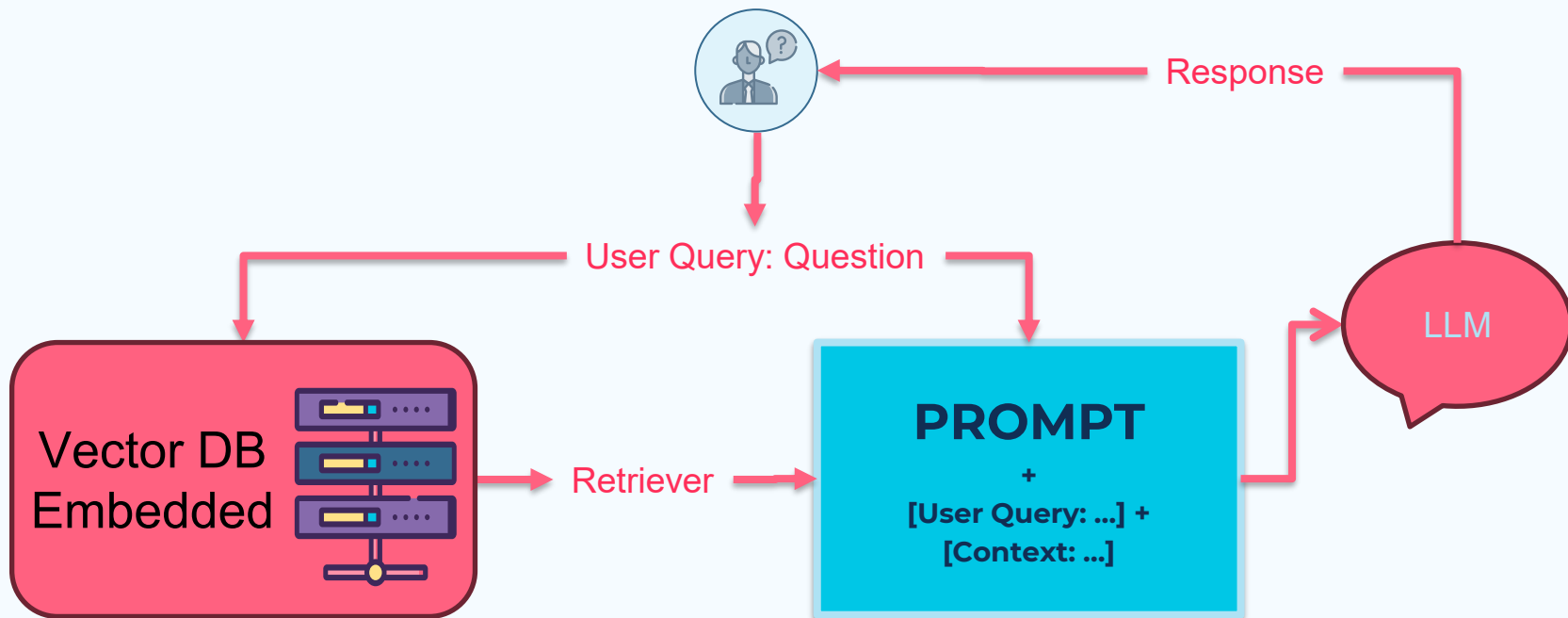
User question is also embedded and then compared to the document.



RETRIEVE

The system searches the database, pulls out the exact text chunks that match the user's question, and appends them to the prompt as verified context for the LLM.

Answer Generation Process



LIVE DEMO

We are using Gemini API key, and Gemini 2.5 model.



What we saw!

What is RAG?

Defination of RAG and why need RAG and how it is useful.

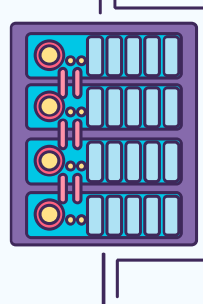
RAG Pipeline

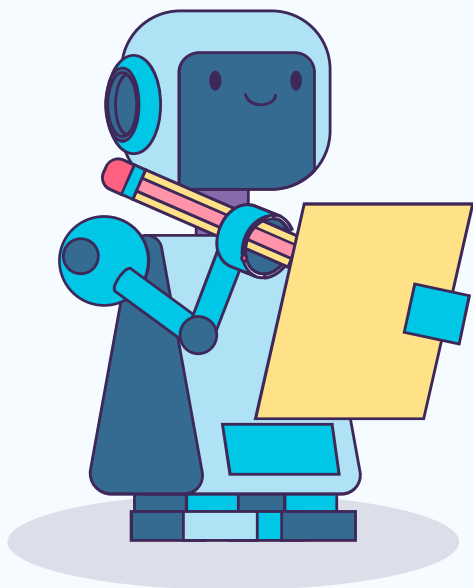
- Doc. injection
- Info. retrival
- Resp. generation

What we learned?

How RAG works and key steps for the RAG

LIVE DEMO!!!





Thanks!

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Alternative resources

Here's an assortment of alternative resources whose style fits the one of this template:

RAG Youtube Video

- https://youtu.be/rS_ksLkhlrM?si=bJQAsGTOqaPruUe7



HTTPS
SSL

